

EXERCISES

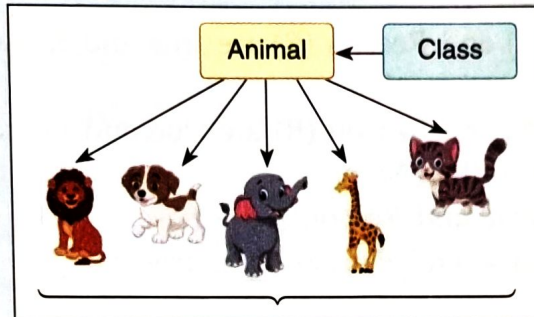


PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions:

Tick(✓) the correct answer:

1.



Name the feature of class in Java depicted in the above picture.

- (a) Displaying a set of animals (b) Class in an object factory
 - (c) Animal is an object factory (d) Object is a factory
2. A class is a template for its:
- (a) data (b) functions (c) objects (d) methods
3. An object is an instance of a:
- (a) function (b) state (c) data (d) class
4. Which is a valid display() method declaration of a class Sample?
- (a) Void Sample{int x} (b) public void display()
 - (c) public void Show() (d) Display{ }
5. Creating an object of a class is also referred to as:
- (a) Initialisation (b) Creation (c) Formation (d) Instantiation
6. By default, a class is declared using the access specifier.....
- (a) private (b) public (c) protected (d) secured
7. Which keyword allocates objects in dynamic memory?
- (a) this (b) private (c) new (d) int
8. Read the following text, and choose the correct answer:
- They are the real world or software world entities which possess different characteristics and common behaviour described within the class.
- Which of the following concept of Object Oriented Programming describe?
- (a) Class (b) Object (c) Quality (d) Category

II. Fill in the blanks:

- 1. A is also considered as an object factory.
- 2. The of a class differ on various characteristics.
- 3. The term is used for creating various objects.
- 4. The class is a for the objects.
- 5. The keyword indicates an operator for dynamic allocation of an object.
- 6. Different objects of a class have common

III. Assertion and Reasoning based questions:

Assertion (A): A car can be defined as a real world object that possesses some characteristics and behaviour.

Reason (R): Each real world object may contain some characteristics but not behaviour.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

IV. Application based questions:

Your friend has tried to write a program using a class as shown below:

```
..... class Friend_list // Line 1
{
    void disp_name( )
    {
        System.out.println("Amit Kumar");
    }
    void disp_address( )
    {
        System.out.println("Lane B: 3/1, Akash Ganga");
        System.out.println("Mayur Vihar, New Delhi");
    }
    public static void main( )
    {
        Friend_list myself = new .....( ); // Line 2
        _____.disp_name( ); // Line 3
        myself. _____.( ); // Line 4
    }
}
```

Due to confusion, he could not complete the above program. He left blank spaces in some of the statements marked with **Line 1, Line 2, Line 3 and Line 4** that are to be filled with appropriate keywords/objects.

Help your friend to complete his program by answering the following questions:

- (a) Name the access specifier that should be filled in the statement marked with **Line 1**.
- (b) Name the element to be filled in the statement marked with **Line 2**.
- (c) What should be filled in the statement marked with **Line 3**?
- (d) What function name should be filled in the statement marked with **Line 4**?

PART B – SUBJECTIVE QUESTIONS

I. Answer the following questions:

1. Define an Object with an example. [ICSE-2006]
2. Mention five states (characteristics) and two methods for the following Classes:
 - (a) class Employee
 - (b) class Bank
 - (c) class School
 - (d) class Book
 - (e) class Park
 - (f) class Medicine
 - (g) class Computer
 - (h) class Camera
3. What is an Object? Give five examples of real world objects.
4. How will you define a software object?
5. Class and Objects are inter-related. Explain.
6. A class is also referred to as 'Object Factory'. Comment. [ICSE-2009]
7. What does the following statement mean? [ICSE-2008]
Employee staff = new Employee();
8. Why is an Object called an 'Instance' of a class? Explain. [ICSE-2010]
9. Why is a class known as a composite data type? [ICSE-2009]
10. Write a statement to create an object 'Keyboard' of the class 'Computer'.
11. Consider a real world object as 'Cricket Ball'. Now, mention two behaviour and methods each by taking the 'Cricket Ball' as a software Object.
12. Refer a class structure as shown below:

```
class MySchool
{
    Name
    Address
    Principal's name
    AcceptData( );
    PrintData( );
}
```

With reference to the above class declaration, indicate whether the following statements are True/False:

- (i) Acceptdata() is the characteristic of the class.
- (ii) Address is behaviour of the class.
- (iii) Acceptdata() and PrintData() are the common behaviour for the objects of class 'MySchool'.
- (iv) Behaviour is the medium of inter-object communication.
- (v) Creating multiple objects of the class 'MySchool' is not possible.

13. You want to create a class 'Football'. Choose the elements to be used as characteristics and behaviour from the list given below:

Ball, Goalkeeper, Making a goal, Defender, Forward player, Passing ball, Referee, Hitting corner, Making fault

Characteristics	Behaviours

14. Write a program by using a class 'Picnic' without any data members but having only functions as per the specifications given below:

class name : Picnic

void display1(): To display venue, place and reporting time.

void display2(): To display number of students, name of the teacher accompanying and bus number.

Write a main class to create an object of the class 'Picnic' and call the functions display1() and display2() to enable the task.